MA 587 Homework 2

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Question 1 Error bound

1(a) Consider a single cell of the problem domain, $L_j = [x_{j-1}, x_j]$, and choose some $z \in (x_{j-1}, x_j)$. The quadratic polynomial that interpolates the points $\{u(x_{j-1}), u(z), u(x_j)\}$ is given by

$$\hat{u}_z(x) = u_{j-1} + \frac{u_j - u_{j-1}}{h} x + A_z (x - x_j)(x - x_{j-1}), \tag{1}$$

where A_z is some constant dependent on the choice of z, $u_j = u(x_j)$, and $h = x_j - x_{j-1}$. This can also be rewritten as

$$\hat{u}_z(x) = \tilde{u}(x) + A_z (x - x_j)(x - x_{j-1}), \tag{2}$$

where $\tilde{u}$ is the linear interpolation of u. Then the error $g(x) = \hat{u}(x) - u(x)$ is zero in at least three points: $g(x_{j-1}) = g(z) = g(x_j) = 0$. By Rolle's theorem, since g has three zeros, there are two points $\xi_{1,z} \in (x_{j-1}, z)$ and $\xi_{2,z} \in (z, x_j)$, for which $g'(\xi_{1,z}) = g'(\xi_{2,z}) = 0$. Applying Rolle's theorem once more, there is another point $\xi_z \in (\xi_{1,z}, \xi_{2,z})$ such that $g''(\xi_z) = 0$. That is,

$$g''(\xi_z) = \hat{u}''(\xi_z) - u''(\xi_z), \tag{3}$$

so, using Equation 1,

$$u''(\xi_z) = 2A_z \tag{4}$$

$$\frac{u''(\xi_z)}{2} = A_z. \tag{5}$$

This value is bounded by

$$\left|\frac{u''(\xi_z)}{2}\right| \leq \max_{a\in[x_{j-1},x_j]}\left|\frac{u''(a)}{2}\right| \leq \max_{y\in[0,1]}\left|\frac{u''(y)}{2}\right| \tag{6}$$

Then, for any $x \in [x_{j-1}, x_j]$,

$$u(x) = \tilde{u}(x) + A_x(x-x_j)(x-x_{j-1}) \tag{7}$$

$$u(x) = \tilde{u}(x) + \frac{u''(\xi_x)}{2}(x-x_j)(x-x_{j-1}) \tag{8}$$

$$u(x) - \tilde{u}(x) = \frac{u''(\xi_x)}{2}(x-x_j)(x-x_{j-1}), \tag{9}$$

so

$$|u(x) - \tilde{u}(x)| = \left|\frac{u''(\xi_x)}{2}(x-x_j)(x-x_{j-1})\right|. \tag{10}$$

By submultaplicativity,

$$|u(x) - \tilde{u}(x)| \leq \left|\frac{u''(\xi_x)}{2}\right| \cdot |(x-x_j)(x-x_{j-1})|. \tag{11}$$

Applying the triangle inequality to the $(x-x_j)(x-x_{j-1})$ term and applying Equation 6 to the $\frac{u''}{2}$ term, this is bounded by

$$|u(x) - \tilde{u}(x)| \leq \max_{y\in[0,1]}\left|\frac{u''(y)}{2}\right| \cdot (|x-x_j| + |x-x_{j-1}|), \tag{12}$$

or

$$|u(x) - \tilde{u}(x)| \leq \max_{y\in[0,1]}\left|\frac{u''(y)}{2}\right| \cdot h \tag{13}$$

$$|u(x) - \tilde{u}(x)| \leq \max_{y\in[0,1]}|u''(y)| \cdot h. \tag{14}$$

1(b) The inequality given is

$$\|(u-u_h)'\| \leq h \cdot \max_{y\in[0,1]}|u''(y)|. \tag{15}$$

The Cauchy-Schwarz Inequality states that $|\langle \boldsymbol{x}, \boldsymbol{y}\rangle| \leq \|\boldsymbol{x}\| \cdot \|\boldsymbol{y}\|$, where $\langle \boldsymbol{x}, \boldsymbol{y}\rangle = \int_0^1 \boldsymbol{x}\boldsymbol{y}dx$ is the inner product, and the norm $\|\cdot\|$ is $\langle \cdot, \cdot\rangle^{1/2}$ Let $\boldsymbol{x} = (u-u_h)'$ and let $\boldsymbol{y} = 1$:

$$|\langle (u-u_h)', 1\rangle| \leq \|(u-u_h)'\| \cdot \|1\| \leq h \cdot \max_{y\in[0,1]} |u''(y)|, \tag{16}$$

and, expanding the inner product definition,

$$\left|\int_0^1 (u-u_h)' dx\right| \leq h \cdot \max_{y\in[0,1]} |u''(y)|. \tag{17}$$

Using a change of variables,

$$\left|\int_0^x (u-u_h)'(s) ds\right| \leq \left|\int_0^1 (u-u_h)'(s) ds\right| \leq h \cdot \max_{y\in[0,1]} |u''(y)| \tag{18}$$

for $x \in [0,1]$, which becomes

$$|[u(x)-u_h(x)] - \cancel{[u(0)-u_h(0)]}| \leq h \cdot \max_{y\in[0,1]} |u''(y)| \tag{19}$$

$$|u(x)-u_h(x)| \leq h \cdot \max_{y\in[0,1]} |u''(y)|. \tag{20}$$

Question 2 Formulations

2(a) First, to show that (V) $\implies$ (M):
Let u^v be a solution to (V), so

$$a(u^v, v) = L(v) \forall v \in \mathbb{V}. \tag{21}$$

Then let $w \in \mathbb{V}$, and consider

$$F(u^v + w) = \frac{1}{2} a(u^v + w, u^v + w) - L(u^v + w). \tag{22}$$

Since a is bilinear and L is linear, this can be rewritten

$$F(u^v + w) = \frac{1}{2} a(u^v, u^v) + a(u^v, w) + \frac{1}{2} a(w,w) - L(u^v) - L(w) \tag{23}$$

$$F(u^v + w) = \frac{1}{2} a(u^v, u^v) + \frac{1}{2} a(w,w) + [a(u^v, w) - L(w)] - L(u^v). \tag{24}$$

From Equation 21, the bracketed term is zero, and

$$F\left(u^v + w\right) = \frac{1}{2}a(u^v, u^v) + \frac{1}{2}a(w, w) - L\left(u^v\right) \tag{25}$$

$$F\left(u^v + w\right) = F(u^v) + \frac{1}{2}a(w, w) \geq F(u^v). \tag{26}$$

Since the bilinear $a(w, w)$ is strictly non-negative, this inequality applies. For any w, $F(u^v) \leq F(u^v + w)$, so for any v, $F(u^v) \leq F(v)$ and (M) is satistified, and thus (V) $\implies$ (M).

Next, to show that (M) $\implies$ (V), consider u^m that satisfies (M). Then u^m minimizes $F()$. Consider the scalar function

$$f(\epsilon) = F(u^m + \epsilon v) \tag{27}$$

for some v. Since u_m minimizes F, then $\epsilon = 0$ minimizes f, so

$$\left.\frac{\partial f}{\partial \epsilon}\right|_{\epsilon=0} = 0. \tag{28}$$

Expanding this yields

$$\left.\frac{\partial}{\partial \epsilon}\left(F(u^m + \epsilon v)\right)\right|_{\epsilon=0} = 0 \tag{29}$$

$$\left.\frac{\partial}{\partial \epsilon}\left(\frac{1}{2}a(u^m + \epsilon v, u^m + \epsilon v) - L(u^m + \epsilon v)\right)\right|_{\epsilon=0} = 0 \tag{30}$$

$$\left.\frac{\partial}{\partial \epsilon}\left(\frac{1}{2}a(u^m, u^m) + a(u^m, \epsilon v) + \frac{1}{2}a(\epsilon v, \epsilon v) - L(u^m) - L(\epsilon v)\right)\right|_{\epsilon=0} = 0 \tag{31}$$

$$\left.\frac{\partial}{\partial \epsilon}\left(\frac{1}{2}a(u^m, u^m) + \epsilon a(u^m, v) + \frac{\epsilon^2}{2}a(v, v) - L(u^m) - \epsilon L(v)\right)\right|_{\epsilon=0} = 0 \tag{32}$$

$$\left. a(u^m, v) + \epsilon a(v, v) - L(v)\right|_{\epsilon=0} = 0 \tag{33}$$

$$a(u^m, v) - L(v) = 0 \tag{34}$$

since this applies for any $v \in \mathbb{V}$, (V) is satisfied and thus (M) $\implies$ (V). Since (M) $\implies$ (V) and (V) $\implies$ (M), (V) $\iff$ (M).

2(b) Since $\mathbb{V}_h$ is a Hilbert space and a is symmetric, linear, and coercive, and from part a, (V) $\iff$ (M), so the variational solution u_h also satisfies

$$F(u_h) \leq F(v) \quad \forall\, v \in \mathbb{V}_h \tag{35}$$
$$2F(u_h) \leq 2F(v) \tag{36}$$
$$a(u,u) + 2F(u_h) \leq a(u,u) + 2F(v) \tag{37}$$
$$a(u,u) + a(u_h,u_h) - 2L(u_h) \leq a(u,u) + a(v,v) - 2L(v). \tag{38}$$

Since u satisfies (V), $a(u,w) = L(w)$ for any w in $\mathbb{V}$, including members of the subset $\mathbb{V}_h$, so

$$a(u,u) + a(u_h,u_h) - 2a(u,u_h) \leq a(u,u) + a(v,v) - 2a(u,v) \quad \forall\, v \in \mathbb{V}_h. \tag{39}$$

This can be rewritten, using the bilinearity of a, as

$$a(u-u_h, u-u_h) \leq a(u-v, u-v) \tag{40}$$
$$\|u-u_h\|_a^2 \leq \|u-v\|_a^2 \tag{41}$$
$$\|u-u_h\|_a \leq \|u-v\|_a \quad \forall\, v \in \mathbb{V}_h. \tag{42}$$

In other words, u_h is the best approximation of u of any function in the discretized set $\mathbb{V}_h$, using a-norm error.

Question 3 Basis Functions

3(a) The three points form a triangle as long as they are not collinear, or if

$$[N_1 - N_2]^T [N_1 - N_3] \neq |N_1 - N_2| \cdot |N_1 - N_3|. \tag{43}$$

3(b) Denoting the basis functions using

$$\phi_j(x,y) = a_j x + b_j y + c_j, \tag{44}$$

the condition can be notated as a linear system

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & h & 0 \\ 1 & 0 & h \end{bmatrix} \begin{bmatrix} c_1 & c_2 & c_3 \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{45}$$

Then the coefficients are simply the entries of

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & h & 0 \\ 1 & 0 & h \end{bmatrix}. \tag{46}$$

This matrix A^{-1} is easily found, since A is lower triangular, to be

$$A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ \frac{-1}{h} & \frac{1}{h} & 0 \\ \frac{-1}{h} & 0 & \frac{1}{h} \end{bmatrix}. \tag{47}$$

Then the basis functions are

$$\phi_1(x, y) = 1 - \frac{x}{h} + \frac{y}{h} \tag{48}$$

$$\phi_2(x, y) = \frac{x}{h} \tag{49}$$

$$\phi_3(x, y) = \frac{-1}{h} + \frac{y}{h}. \tag{50}$$

Since a 2d linear function can be represented using coefficients a, b, c, any basis that spans $\mathbb{R}^3$ can be a basis for 2d linear functions. The columns of A^{-1} are mutually orthogonal, and do span $\mathbb{R}^3$, so these basis functions ϕ_j do form a basis for 2d linear functions.

3(c) First, the gradients of the basis functions are

$$\nabla\phi_1(x, y) = \begin{bmatrix} \frac{-1}{h} & \frac{1}{h} \end{bmatrix}^T \tag{51}$$

$$\nabla\phi_2(x, y) = \begin{bmatrix} \frac{1}{h} & 0 \end{bmatrix}^T \tag{52}$$

$$\nabla\phi_3(x, y) = \begin{bmatrix} 0 & \frac{1}{h} \end{bmatrix}^T. \tag{53}$$

Then, the stiffness matrix entries are

$$a(\phi_1, \phi_1) = \int_K \nabla\phi_1 \cdot \nabla\phi_1 dx \tag{54}$$

$$= \int_K \left(\frac{1}{h}\right)^2 + \left(\frac{1}{h}\right)^2 dx \tag{55}$$

$$= \frac{2}{h^2} \int_K dx \tag{56}$$

$$= 1 \tag{57}$$

$$
\begin{aligned}
a(\phi_1, \phi_2) &= \int_K \nabla\phi_1 \cdot \nabla\phi_2 dx && (58)\\
&= \int_K \left(\frac{1}{h} \cdot \frac{-1}{h}\right) dx && (59)\\
&= \frac{-1}{h^2} \int_K dx && (60)\\
&= \frac{-1}{2} && (61)
\end{aligned}
$$

$$
\begin{aligned}
a(\phi_1, \phi_3) &= \int_K \nabla\phi_1 \cdot \nabla\phi_3 dx && (62)\\
&= \int_K \left(\frac{1}{h} \cdot \frac{-1}{h}\right) dx && (63)\\
&= \frac{-1}{h^2} \int_K dx && (64)\\
&= \frac{-1}{2} && (65)
\end{aligned}
$$

$$
\begin{aligned}
a(\phi_2, \phi_2) &= \int_K \nabla\phi_2 \cdot \nabla\phi_2 dx && (66)\\
&= \int_K \left(\frac{1}{h} \cdot \frac{1}{h}\right) dx && (67)\\
&= \frac{1}{h^2} \int_K dx && (68)\\
&= \frac{1}{2} && (69)
\end{aligned}
$$

$$
\begin{aligned}
a(\phi_2, \phi_3) &= \int_K \nabla\phi_2 \cdot \nabla\phi_3 dx && (70)\\
&= \int_K 0 dx && (71)\\
&= 0 && (72)
\end{aligned}
$$

$$
\begin{aligned}
a(\phi_3, \phi_3) &= \int_K \nabla \phi_3 \cdot \nabla \phi_3 dx && (73)\\
&= \int_K \left(\frac{1}{h} \cdot \frac{1}{h}\right) dx && (74)\\
&= \frac{1}{h^2} \int_K dx && (75)\\
&= \frac{1}{2}, && (76)
\end{aligned}
$$

The stiffness matrix is symmetric since a is bilinear, so the stiffness matrix A matches the one given in the problem description.

Question 4 Sobolev Space

4(a)

$$
\begin{aligned}
\|u\|_{H^k} &= \sqrt{(u,u)_{H^k}} && (77)\\
\|u\|_{H^k}^2 &= (u,u)_{H^k} && (78)\\
&= \sum_{|\alpha| \leq k} \langle D^\alpha u, D^\alpha u \rangle_{L_2(\Omega)} && (79)\\
&= \sum_{|\alpha| \leq s} \langle D^\alpha u, D^\alpha u \rangle_{L_2(\Omega)} + \sum_{s < |\alpha| \leq k} \langle D^\alpha u, D^\alpha u \rangle_{L_2(\Omega)} && (80)\\
&= \|u\|_{H^s}^2 + \sum_{s < |\alpha| \leq k} \langle D^\alpha u, D^\alpha u \rangle_{L_2(\Omega)}. && (81)
\end{aligned}
$$

Since the L_2 inner product is strictly non-negative, the rightmost term in Equation 81 is nonnegative, so

$$
\begin{aligned}
\|u\|_{H^s}^2 + \sum_{s < |\alpha| \leq k} \langle D^\alpha u, D^\alpha u \rangle_{L_2(\Omega)} &= \|u\|_{H^k}^2 && (82)\\
\sqrt{\|u\|_{H^s}^2 + \sum_{s < |\alpha| \leq k} \langle D^\alpha u, D^\alpha u \rangle_{L_2(\Omega)}} &= \|u\|_{H^k} && (83)\\
\|u\|_{H^s}^2 &\leq \|u\|_{H^k} && (84)
\end{aligned}
$$

4(b)

$$\|u+v\|_{H^k}^2 = (u+v, u+v)_{H^k} \tag{85}$$

$$= (u,u)_{H^k} + (v,v)_{H^k} + 2\,(u,v)_{H^k} \tag{86}$$

$$\|u+v\|_{H^k}^2 = \|u\|_{H^k}^2 + \|v\|_{H^k}^2 + 2\sum_{|\alpha|\le k} \langle D^\alpha u, D^\alpha v\rangle_{L_2(\Omega)}\,. \tag{87}$$

By Cauchy-Schwarz in the L_2 norm,

$$\|u+v\|_{H^k}^2 \le \|u\|_{H^k}^2 + \|v\|_{H^k}^2 + 2\sum_{|\alpha|\le k} \|D^\alpha u\|_{L_2(\Omega)} \|D^\alpha v\|_{L_2(\Omega)} \tag{88}$$

$$\|u+v\|_{H^k}^2 \le \|u\|_{H^k}^2 + \|v\|_{H^k}^2 + 2\sqrt{\left(\sum_{|\alpha|\le k} \|D^\alpha u\|_{L_2(\Omega)} \|D^\alpha v\|_{L_2(\Omega)}\right)^2}\,. \tag{89}$$

Again by Cauchy-Schwarz,

$$\|u+v\|_{H^k}^2 \le \|u\|_{H^k}^2 + \|v\|_{H^k}^2 + 2\sqrt{\left(\sum_{|\alpha|\le k} \|D^\alpha u\|_{L_2(\Omega)}^2\right)\left(\sum_{|\alpha|\le k} \|D^\alpha V\|_{L_2(\Omega)}^2\right)} \tag{90}$$

$$\le \|u\|_{H^k}^2 + \|v\|_{H^k}^2 + 2\,\|u\|_{H^k}\cdot\|v\|_{H^k}\,, \tag{91}$$

Taking the square root, this becomes

$$\|u+v\|_{H^k} \le \|u\|_{H^k} + \|v\|_{H^k} \tag{92}$$